Samaranjan Manjhi

C/C++ Systems & Cyber Security Software Developer

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SUMMARY

C/C++ Systems and Cyber Security Engineer with **3+ years** of experience building production-grade endpoint security tools, kernel monitors, binary integrity frameworks, and Data Loss Prevention (DLP) systems for **macOS and Linux**. Deep expertise in Endpoint Security Framework (ESF), execution gating, system call hooking, Qt GUI development, and cross-platform C++ architecture. Delivered **10+ enterprise security modules** deployed globally across macOS and Linux endpoints for enterprise antivirus suites.

TECHNICAL SKILLS

Languages: C, C++, Bash Scripting, Relational SQL
Frameworks: Endpoint Security Framework (ESF), Extended Berkeley Packet Filter (eBPF), System Audit, Core Foundation
Security: Cyber Security Architecture, Sodium Cryptography Library, OpenSSL Library
Tools: Git, CMake, Make, Xcode, GDB, Valgrind, KDB/KGDB, SQLite, Boost, Nano
Core CS: Data Structures, Multithreading, Inter-Process Communication, TCP/IP Networking, OS Internals (Linux/macOS)

PROFESSIONAL EXPERIENCE

Systems & Security Software Engineer | MicroWorld Software Services Pvt. Ltd. — Mumbai

Aug 2023 – Present

- Designed and delivered **10+ production security modules** spanning kernel monitors, DLP engines, binary integrity systems, firewall rule generators, and cloud backup agents — deployed globally across enterprise Linux and macOS systems running security products.
- Constructed high-performance cross-platform security daemons in C/C++, integrating Linux kernel modules, macOS ESF event loops, and POSIX Unix domain socket IPC to reduce system memory overhead by 25% and improve event detection throughput by 40%.
- Independently owned end-to-end design, implementation, and deployment for select modules, coordinating with QA and product teams to define requirements and validate detection accuracy prior to release.
- Regularly mentored and onboarded new engineers on secure C/C++ development, kernel-level debugging, and internal security frameworks, shortening ramp-up time on kernel and DLP module development.

KEY PROJECTS

Real-Time Virus Scanning — eBPF | C/C++, eBPF, Kernel Tracing, AV Engine Integration

Mar 2026 – Jun 2026

- Built a real-time on-access antivirus scanning daemon using eBPF programs attached to kernel hooks to intercept file open, write, and execution events, streaming new and modified files into the AV scan engine for detection.
- Integrated quarantine and process-termination actions triggered on malware detection, with a configurable watchlist and root/parent-process validation to reduce false positives and redundant re-scans.
- Optimized event throughput with a lock-free audit map and dedicated scanner thread pool, sustaining real-time on-access detection under heavy file I/O without blocking user processes.

MW Guardian — Binary Integrity & Execution Gating | C/C++, Linux Kernel, Cryptographic Verification

Feb 2026 – Jul 2026

- Enforced execution gating via **kernel permission events** to intercept binary execution at kernel level, synchronously validating cryptographic signatures and terminating untrusted binaries prior to execution.
- Architected a multi-key cryptographic signing pipeline supporting key rotation while retaining legacy public keys to preserve signature validity for previously signed binaries.
- Created startup self-verification routines where guardian and verifier components validate their own binary signatures prior to initiating the main event loop, eliminating bootstrap trust gaps.

Two-Factor Authentication for SSH | C/C++, PAM Module, TOTP Cryptography, Qt Framework, macOS

Jul 2026 – Aug 2026

- Extended the PAM authentication stack with a custom TOTP verification module, layering a second authentication factor onto SSH login while preserving compatibility with existing PAM configuration and login flows.
- Built and integrated a Qt-based 2FA enrollment and management widget directly into the production macOS security console GUI, allowing administrators to enable, configure, and audit 2FA policy per endpoint.
- Implemented secret provisioning and QR-code based device enrollment, coordinating configuration state between the PAM module and GUI layer without requiring a service restart.

Printer Control & Data Loss Prevention (DLP) | macOS, C/C++, ESF, Print Service Libraries, Xcode

Oct 2025 – Jan 2026

- Developed a macOS DLP driver utilizing ESF and print spool monitoring to inspect print files for sensitive information, cancelling unauthorized print jobs prior to printing.
- Applied application-level process whitelisting and blacklisting enforced via ESF process identity, bypassing clean content for trusted applications and blocking unauthorized applications outright.
- Archived scanned print jobs alongside result metadata to maintain audit trails for compliance reporting and post-event forensic investigations.

File Integrity Monitoring (FIM) | macOS, C/C++, File System Events, SQLite Database, CommonCrypto MD5

Jun 2025 – Aug 2025

- Built a file system event daemon storing cryptographic file hashes, permissions, file sizes, and timestamps in SQLite, emitting real-time security alerts upon unauthorized path modifications, renames, or deletions.
- Established baseline state snapshots on initial startup and executed diff comparisons on subsequent runs to output structured security event logs across configurable watch paths.

Content Scanner — DLP Engine & CLI | C/C++, Regex Engine, Optical Character Recognition, Archive Libraries

Sep 2024 – May 2025

- Built a multi-threaded DLP scanner supporting **15+ file formats** via regular expression parsing and optical character recognition, packaged as a shared library and CLI utility.
- Developed a high-performance Unix domain socket server variant allowing clients to stream file paths and retrieve scan results without re-initializing regex engines, reducing per-scan latency by 60%.

EDUCATION

Bachelor of Engineering — Computer Engineering

2019 – 2023

Shree L.R. Tiwari College of Engineering, Mumbai University — CGPA: **8.56 / 10**